

STATE OF THE ART IN BIOTECHNOLOGY

Lecture Notes, Day 1

Main point of this session:

Today, we will build a foundation of what bioengineering encompasses, specifically focusing on medical applications

Learning objectives for today:

- Basic concepts in cell biology and genetics
- Innovations in the field of biomedicine

Bioengineering Learning Objectives

- Students finish the course more informed about the state of bioengineering today, and how it has developed over time
- Students are excited about the breadth of possibilities encompassed by genetic engineering and all of the fields which it applies to
- Students understand specific inventions and developments in the field of bioengineering, as well as learn foundational biological knowledge which applies to these inventions

Bioengineering 2-Day Course Schedule

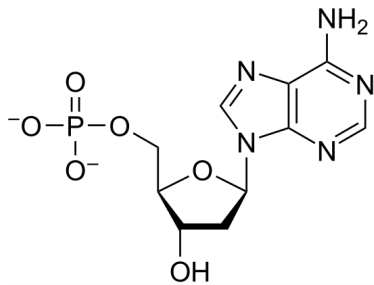
- Day 1A (FIRST HOUR)
 - Understanding Bioengineering
 - Bioengineering uses engineering applications to understand, modify, and control living systems
 - What do Bioengineers do? They study biological processes and integrate them with engineering principles to develop solutions in a wide range of fields. For example:
 - Design medical devices (Glucose sensors, materials and structures for hip-replacements)
 - New processes to clean the Earth's air and water resources
 - Genetically engineer bacteria to produce fuel, pharmaceutical products, or remove toxins
 - Create plants which can withstand climate change and produce larger yields to feed growing populations
 - New materials with edited microbial DNA to enable things like self-repairing fabric
 - Basic biological concepts to cover (These should give a basis upon which to understand the important developments highlighted later):
 - Eukaryotes vs Prokaryotes

- Eukaryotes is most often multicellular organisms, whereas prokaryotes are unicellular
- Animals and plants are eukaryotic, while bacteria are prokaryotic
- Eukaryotic cells are 100-10,000 times larger than prokaryotic cells and are much more complex

■ DNA and Genetics (Nucleic acids, Chromosomes, etc)

• What is DNA?

- Deoxyribonucleic Acid is a molecule carrying genetic information for the functioning and development of an organism
- DNA is made up of two linked strands wound around each other to form a shape called the double helix
- Breaking it down further, DNA is composed of building blocks called nucleotides (A sugar molecule, a phosphate group, and a base)



- The four types of bases: Adenine (A), Cytosine (C), Guanine (G), and Thymine (T)
- Chemical bonds between these bases holds the strands of DNA together
- A bonds with T, and C bonds with G
- The specific sequence of bases in a strand of DNA encodes genes and biological information, such as instructions for making a protein or RNA molecule

• Chromosomes

- Thread-like structures composed of protein and a single molecule of DNA
- Humans have a total of 46 chromosomes
 - Children inherit half of their chromosomes from their mother, and half from their father
- During cell division, chromosomes provide a structure which ensures that DNA is replicated and distributed appropriately to the daughter cells

• Cell division

- The process in which one cell (the parent cell) divides to form two new cells (daughter cells)
- Prokaryotes and Eukaryotes have different processes of cell division
- Prokaryotes split through binary fission, while eukaryotic cells split through mitosis and meiosis
- Before both mitosis and meiosis, the part of the cell cycle called interphase duplicates the cell's chromosomes. This initiates the process of cell division.

■ Mitosis

- Used in division of body cells

- Have four stages: Prophase, Metaphase, Anaphase, and Telophase (PMAT)
 - In mitosis, PMAT occurs once
 -
 - Meiosis
 - Used in division of gamete cells (sperm and egg cells)
 - Have four stages: Prophase, Metaphase, Anaphase, and Telophase
 - In meiosis, PMAT occurs twice
 - Mitosis vs Meiosis video:

<https://www.youtube.com/watch?v=zrKdz93WIVk>
- Stem Cells
 - Cells that can become any other cell
 - i.e. They are the body's raw material for all tissues
 - No other cell has the ability to do this
- DNA damage and repair
 - There are a collection of processes by which an organism can repair its DNA
 - This is covered more in CRISPR subunit, but worth knowing that the ability of a cell to repair its DNA is vital to the normal functioning of an organism
 - Many genes shown to influence life span of an organism are involved in DNA repair and protection
- Epigenetics
 - The study of how cells control gene activity without changing the DNA sequence (i.e. which genes are activated or deactivated)
- RNA
 - Whereas the sugar in DNA is Deoxyribose, the sugar in RNA is ribose
 - RNA has the bases Adenine (A), Uracil (U), Cytosine (C), and Guanine (G)
 - In other words, T is switched to U
 - While DNA stores genetic information, RNA uses that information to make proteins
- Show animation overview from Veritasium video:

https://www.youtube.com/watch?v=X_tYrnv_o6A
- Day 1B (SECOND HOUR)
 - Biomedicine: Let's focus on bioengineering's most well-recognized application
 - Cover several pertinent inventions in medicine today which bioengineers have developed to impact human health
 - Tissue engineering

- Can human organs be produced in a lab or even 3-D printed?
Some people today have bladders produced in a lab from the patients' own cells. Growing more significant organs (hearts, livers, etc) is one of the major objectives in bioengineering, and many fascinating developments have been made. Currently, efforts center around the following process:
- Collecting Cells
 - Engineering a tissue or organ begins with using the right type of cells
 - In some cases, cells are isolated from a tissue sample the size of a postage stamp, then mixed with growth factors and multiplied in a lab
 - These cells multiply so rapidly that, in approximately 6 weeks, a 1-cell-thick layer could theoretically span an entire football field
 - For cell types that cannot be adequately grown outside the body (e.g. heart, nerve, liver, pancreas cells) stem cells may be an option because of their ability to become multiple cell types
- Making a Scaffold
 - After cell expansion, scaffold is created in the shape or mold of the tissue being engineered
 - One technique for certain tissues is electrospinning, a fabrication technique used to produce nano- or micro-diameter fibers to create biocompatible and biodegradable scaffolds
 - Used for blood vessels, muscles, and tendons
 - Involves dissolving a biomaterial into a solvent, loading it into syringe, and applying high voltage to the biomaterial as it is gradually ejected from the syringe
 - At some distance away, there is a metal collection device which is grounded so that it attracts the charged solution when leaving the syringe. Ultimately the solution forms a tubular structure for use as a mold.
 - Bioprinting:
 - The active cells are deposited in the biodegradable scaffolds in controlled conditions
 - similar to 3d printing, but uses bio-inks, consisting of human cells instead of plastics
 - Cells are printed in thin layers that accumulate into living tissue or body parts that can be implanted
 - Skin Printing

- Eventually, a bioprinter with a patient's own cells can be brought to their bedside to treat large wounds or burns by printing skin, layer by layer, to begin the healing process
 - researchers have demonstrated proof-of-concept of the system by printing skin directly onto pre-clinical models
 - the next step is to conduct a clinical trial with humans
 - Testing Organ
 - Organ baths are used to test the organ. It is not enough for the cells to simply look like the organ, they must also function
 - Organ baths are where tissues are suspended to test how they contract and react to chemical and electrical impulses, to see if their responses are normal
 - CRISPR/Cas9
 - CRISPR stands for Clustered Regularly Interspaced Short Palindromic Repeats, and it was previously known as an antiviral mechanism bacterial species use. The Nobel prize was awarded to Emmanuelle Charpentier and Jennifer Doudna, who made key discoveries in DNA manipulation to discover the CRISPR-Cas9 gene editing system.
 - Not the only gene-editing method, but the fastest and most efficient
 - Most pre-CRISPR medical therapies for serious illnesses aimed to treat patients by targeting abnormal proteins associated with the disease
 - Usually these abnormal proteins exist because of mutations in specific genes in the patient's DNA. In the past we have not been able to alter these genes directly to address root cause of disease
 - However, the gene editing system known as CRISPR/Cas9 allows us to modify, delete, or correct specific areas of our DNA in order to treat these diseases
 - How does it work?
 - The system is comprised of 2 parts:
 - Cas9, a nuclease (enzyme that cuts DNA)
 - A segment of RNA containing a "guide" sequence directs Cas9 to the specific location in the DNA where the edit should be made
 - Cas9 associates with the guide RNA to form a complex that can be easily and precisely targeted to the desired site in the DNA
 - The gene editing process begins when the complex recognizes and binds to a short segment of DNA adjacent to the target site.
The "guide" RNA identifies the intended target sequence in

the DNA and the Cas9 enzyme cuts the DNA, forming a double strand break.

- Our cells respond to such breaks by activating natural DNA repair pathways.
- **Deletions**
 - Non-homologous end joining (NHEJ)
 - Means you don't need other bits of DNA to act as a template for repairs
 - All the tools needed to do this come naturally in cells. In fact, all cells (bacterial and eukaryotic) are able to repair breaks in DNA on their own, because it happens actually quite commonly
 - In any particular cell, there will be between 10 and 50 double strand breaks in the DNA per day. A significant amount!
 - Cells need to be able to fix those breaks, and one of the ways they do that is by NHEJ
 - Here's how it works: A strand of DNA is broken; it might have been an accidental break OR it may have been cut by the Cas9 enzyme
 - The cell does not distinguish between the two, and will go about repairing it
 - For a deletion, two breaks may be made at either end of a sequence, cutting the intervening section out.
 - One might wonder, if the DNA is repaired, then what was the point of CRISPR causing the break?
 - Although the DNA is rejoined, it is often not joined perfectly. Process is error-prone, sometimes a nucleotide is lost, sometimes an extra nucleotide is incorporated in the DNA.
 - Scientists use this technology to hopefully induce a mutation in DNA that might knock out a gene
 - Knocking out a gene means the gene has been prevented from working, or has become non-functional
 - Imagine a sequence that went UGC, and you caused a break after the G. It's possible an extra nucleotide might be added there, and if it is an Adenine, the sequence now reads UGA which is a STOP codon.

Prevents gene from being transcribed and translated into a protein

- **Insertions**

- Corrections to DNA can also be made by adding a DNA template along with a Cas9 guide RNA complex.
- Homology Directed Repair (HDR)
 - Another natural repair mechanism for double strand DNA breaks, but one much less error-prone than NHEJ
 - Broken DNA is rejoined using a template of nucleotides which match the intended repair
- Following a break from the Cas9 enzyme, the targeted site is also provided with a donor template containing a desired insertion or modification, flanked by segments of DNA homologous to the two ends of broken DNA.
 - The cell uses the template to repair the break, thereby replacing the faulty DNA sequence or inserting a novel gene
- Although this technology emerged in 2012, it is only beginning to be used in human subjects
 - Although promising, there are concerns
 - A study found that CRISPR applied to human embryo cells sometimes prompted cells to lose entire chromosomes. Administered to repair a gene for hereditary blindness, the CRISPR-Cas9 technology appeared to wreak genetic havoc in half the specimens which researchers tested, according to the [study](#) (published in the journal "Cell")
 - It is proposed that cells can be so confused by large DNA alterations that they give up on fixing them and remove entire chromosomes as a result
- PAM Sequence
 - Editing genomes with CRISPR proteins involves a sequence called PAM
 - It all starts with what CRISPR systems originally evolved to do: defend bacteria against virus
 - bacteria face the constant threat of infection and destruction by a special type of virus: the bacteriophage
 - IF bacteriophages are able to inject their DNA genome into the bacterial cell, replicate inside, and burst out, the cell is destroyed. In response, bacteria have evolved a protective immune system: CRISPR
 - CRISPR array is a short stretch of DNA in bacteria, composed of alternating repeated sequences and target-specific spacers (DRAW IT OR SHOW DIAGRAM).

These spacers contain the DNA of invading viruses collected from past infections

- When a virus infects the bacterium, a new space is added into the growing CRISPR array. This process begins when a protein complex (Cas1, and Cas2) identifies the invading viral DNA and cuts out a segment of DNA, known as the protospacer.
- A Protospacer is inserted into the front of the CRISPR array. Now bacteria has an embedded memory of this phage infection. If phage returns, bacteria is armed to defend itself
 - This defense starts with transcribing a long CRISPR RNA (pre-crRNA) from the repeats and spacers of the CRISPR array. Another RNA called the tracer RNA comes along and links up with the CRISPR RNA through base-pairing. A protein known as Cas9 grabs onto the two RNA and trims it to a more manageable size to form a complete search complex
 - If the sequence of the crRNA matches the sequence of the invading virus, Cas9 cuts the viral DNA and destroys the phage, allowing the bacterial cell to survive.
 - The viral DNA that is targeted by the search complex, is the exact same sequence as the DNA in the crRNA. So how is Cas9 able to distinguish between itself and the enemy? This is where the Protospacer Adjacent Motif comes in, a specific sequence of nucleotides around 2-6 base pairs that follows the protospacer sequence in the viral genome.
 - In *Streptococcus pyogenes*, Cas9 recognizes the PAM sequence GG. This PAM sequence must be present for the Cas9 protein to know that it is ok to latch onto and cut this region of DNA. Basically tells Cas9 that it is an enemy.
 - The spacer sequences within the CRISPR array are not followed by a GG, but GTT. This means the Cas9 is unable to bind to the CRISPR array, thereby avoiding cutting the bacterium's own genome.
 - How is there always a PAM sequence at a true Cas9 target? In the DNA acquisition step we

covered previously, we mentioned that the Cas1Cas2 protein complex is in charge of capturing new spaces from incoming viral DNA

- Cas9 works with Cas1 and Cas2 to find the PAM sequence and remove the protospacer next to it
 - Only picking targets with PAMS guarantees that when the same virus infects again, and Cas9 is armed with the matching CRISPR RNA guide, it will be able to destroy the viral DNA
- PAM sequence also serves an additional role
 - searching through all the dna inside a bacterial cell would take a very long time, but the PAM sequence accelerates the search process
 - Instead of trying to unwind every bit of DNA to find a match, Cas9 bounces around the cell in search of a tiny PAM sequence. If it finds one, only then does it check to see if the CRISPR RNA matches
 - So how is the PAM sequence involved in Cas9 genome editing?
 - If scientists want to use Cas9 to cut human DNA, or the DNA of any other organism, they first look for a PAM sequence within the target genome and then design an RNA to match the sequence next to it.
 - What if there is no GG next to what scientists want to cut? Fortunately, GG is not the only PAM. Scientists can edit genes with Cas9s from different organisms and even different CRISPR proteins. These different Cas9s have all evolved to recognize distinct PAM sequences. Scientists have even engineered the S. pyogenes Cas9 to recognize other PAMS

- Day 2A (FIRST 1.5 HOURS)
 - Agri-food
 - Population dynamics, land availability, and climate change are all variables in an increasingly complex and interdependent global agriculture industry
 - World's population is expected to grow to nearly 10 billion by 2050, increasing agricultural demand by 50% compared to 2013 levels.
 - With increased consumption, bioengineering has become a major factor in the growth and sustainability of global food supplies
 - CRISPR is leading the way and generating the most headlines in agriculture, and has been used in a variety of applications
 - boosting crop resistance
 - engineering crops to produce their own biological pesticides
 - protecting against disease in livestock
 - Scientists developed a tomato plant which not only grows in briny soil—in fact, in soil with a salt content 50 times higher than tomatoes can typically tolerate—but which actually reduces the salinity of the soil as it is grown, reclaiming fields for re-use.
 - Scientists furthermore expect the responsible gene to work in nearly any plant, which would allow reclamation of fields nearly anywhere on the planet, not just where tomatoes grow
 - Crops engineered to resist specific herbicides
 - allows the creation of “designer” herbicides with no lasting ecological impact
 - These bio-herbicides kill weeds but not the resistant plant, and then biodegrade into harmless components
 - Targeted herbicides, by eliminating weeds, reduce the need to till the soil. This permits topsoil ecosystems to flourish and reduces the need for fertilizers
 - Annually, 25 million acres of agricultural land are lost to high soil salinity, which can be the result of long-term irrigation
- Day 2B Philosophical/Ethical Questions (0.5 Hours)
 - Mary Shelley's *Frankenstein* is often credited with introducing the general public to the moral and ethical questions surrounding genetic engineering
 - In the story, Victor Frankenstein brings to life a humanoid monster, immediately regrets his actions, and abandons the creature. Dr. Frankenstein's actions cause suffering for himself, others and his creation.
 - Frankenstein's monster, doubtless an immense scientific creation, is a warning (whether or not Shelley intended it as such) that science can be a double-edged sword

- While genetic engineering has saved countless lives and benefited society immeasurably, there are always risks of misuse, accidents, and unintended consequences.
 - Accidental release of diseases, GMO pollens, and invasive species must be recognized and are already an issue today.
 - Even from my lab at MIT's Whitehead Institute, despite careful measures to prevent this, *Arabidopsis thaliana* plants (the organism we had been modifying and studying) were found growing between cracks in the pavement outside the building.
 - Especially with diseases, it can be very difficult to 100% prevent accidental release out of a lab
- Example of He Jiankui
 - Chinese scientist imprisoned for his illegal use of genome editing on human embryos
 - Announced he had created the first human genetically edited babies, leading to ethical and legal controversies
 - <https://www.ft.com/content/126689a4-f48e-11e8-9623-d7f9881e729f>

CUT SECTIONS (MAY ADD IF TIME PERMITS):

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- Basics of Cell Biology: Transport and Signaling
 - Understanding how cells control, enhance, and regulate reactions. Foundational understanding of cells helps you in bioengineering, particularly dealing with living organisms such as plants or animals
 - Membranes, Lipids, and Proteins
 - cell has a barrier. How does barrier actually get traversed? Cell cannot have impenetrable shell, it must be able to receive information from and exchange resources with outside
 - Eukaryotic cells are majority of what we'll focus on
 - Think about this barrier, how it is set up and how it is bypassed.
 - Barriers

- Lets cell know the inside from the outside
- In the context of the organelle, the cell can know the difference between the cytosol and the lumen
- Big barrier we often hear about is the plasma membrane, whereas there is also the organelle membrane. Many of the things we talk about regarding plasma membranes can also be talked about in the context of organelle membranes, but there are differences to consider.
- Phospholipid Bilayer: What about oil and water does not allow them to mix (can show oil and water demo)
 - two parts of oil, hydrophilic part, and hydrophobic part. If we have water, oil is organizing itself so hydrophilic parts are facing the water. (shake demo) This ability to self assemble and self organize is the basic property of how our membranes can separate themselves from an aqueous environment
 - When oil groups together using hydrophobic and hydrophilic property, it forms a layer
 - one of the components in the plasma membrane that creates barrier is the phospholipids. It has many of the same properties of oil that allow it to create a barrier against the aqueous environment
 - Phospholipids, Cholesterol, Glycolipids, and of course Proteins
 - Top of lipid has an alcohol group, which is linked to phosphate, which is connected to the glycerol backbone. This is further connected to fatty acids (draw the whole structure)
 - It has polar headgroup, which is hydrophilic, whereas fatty acid tails confer the hydrophobic side
 - BUT both outside and inside of cell are aqueous, so how does this make sense. This is why we have a phospholipid bilayer
 - phospholipids self assemble with polar headgroups sticking out, and hydrophobic core buried inside
 - Phospholipid bilayer does great job preventing diffusion of hydrophilic things, including proteins, sugars, and ions. It is important to consider what can and cannot pass through the layer. It is not a solid brick wall, but can allow things through and facilitate communication between cells

■ Topics

- What are the components of this barrier, and introduce idea of fluidity
- Introduce heterogeneity, and how within these membranes it is actually really important.
 - Charge is why this heterogeneity is important. Also Shape.
 - Alcohol groups
- Asymmetry. Really interesting phenomenon that happens in cells, meaning topology matters
- What would you allow gene editing for? What about making it more intelligent, blond hair?